The Selected Higher-Star Defect of a Design Incidence Lattice

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Abstract

For a t-design D with point-by-block incidence matrix A we study the *selected higher-star defect* $\Delta(D) = H_k/\operatorname{im}_{\mathbb{Z}}(A)$, where $H_k = \{z \in \mathbb{Z}^P : \sum z \equiv 0 \pmod{k}\}$ and k is the block size. With $d = \gcd\{\lambda_r - \lambda_{r+1}\}$ over the star vector, we prove the sandwich $dH_k \le \operatorname{im}_{\mathbb{Z}}(A) \le H_k$, so $\Delta(D)$ is a finite abelian group killed by d, on at most $v-1$ generators, of order dividing d^{v-1}; it is the torsion of the retained-apex cokernel and fits a canonical sequence $0 \to \Delta(D) \to \operatorname{coker}(A) \to \mathbb{Z}/k \to 0$. A pointed presentation identifies $\Delta(D)$ with the cokernel of an explicit $v \times b$ matrix whose Gram matrix depends only on the ordinary design parameters, yielding a square-order bound and a bilinear discriminant form into which the dual defect embeds isotropically. Analyzing that form prime by prime gives: a complete good-prime decomposition and, at odd valuation one, a Witt obstruction that strictly beats square divisibility on an infinite family of $S(2, 5, v)$; an exceptional-prime rank-three reduction; and, at the prime two in higher valuation, a metabolicity theorem showing the growing constituent is always split, so no binary analogue of the odd-prime *discriminant-form* obstruction grows with v (other binary obstructions are not excluded). We give fixed-parameter phenomena (a valuation-two formula, a same-parameter separation of two Steiner triple systems, and the complete STS(15) spectrum), a local Pasch transport law, an elementary realization theorem, and geometric applications. Classical ingredients are attributed throughout; no priority or firstness is claimed. The finite controls are reproduced by an included script.

Contents

1 Introduction **2**

2 The higher-star defect **3**

3 Pointed presentation and the parameter envelope **4**

4 The discriminant form prime by prime **4**

4.1 Good primes 4

4.2 Exceptional primes and the binary boundary 5

5 Fixed parameters: determinacy and its failure **6**

6 Local transport **7**

7 Geometric applications **7**

A The STS(15) defect spectrum **8**

B Realization of elementary defects **8**

C A nonzero example: SQS(8) **9**

D Reproducibility of the finite controls **9**

1 Introduction

Let $D = (P, \mathcal{B})$ be a nonempty, possibly non-simple t-(v, k, λ_t) design with $2 \le t \le k < v$, and let $A \in \mathbb{Z}^{P \times \mathcal{B}}$ be its point-by-block incidence matrix. Write $r = \lambda_1$, $\lambda = \lambda_2$, $\alpha = r - \lambda$, $n = v - 1$, $b = |\mathcal{B}|$, and $L = \mathrm{im}_{\mathbb{Z}}(A) \subseteq \mathbb{Z}^P$ for the integral column (image) lattice. Because every column has weight k, L lies in the "augmentation lattice"

$$H_k = \Big\{ z \in \mathbb{Z}^P : \textstyle\sum_{x \in P} z_x \equiv 0 \pmod{k} \Big\}.$$

Definition 1.1. The *selected higher-star defect* of D is the finite (Section 2) abelian group $\Delta(D) = H_k / L$.

Equivalently, if A^+ is A with an appended all-ones block row, then $\Delta(D)$ is the torsion subgroup of $\mathrm{coker}(A^+)$; it measures the failure of the block columns to generate everything summing to $0 \bmod k$. This paper organizes what can be said about $\Delta(D)$ around four questions: *why is it finite; how is it presented and bounded; which groups occur at fixed parameters; and how does it change under a local design operation.*

Results. Theorem 2.1 bounds $\Delta(D)$ between dH_k and H_k and derives its exact sequences; Proposition 3.1 gives a pointed presentation and a parameter-only Gram/discriminant envelope. We then analyze the discriminant form prime by prime: the good-prime decomposition, the odd valuation-one Witt obstruction (Theorem 4.2), which strictly improves square divisibility on an infinite $S(2, 5, v)$ family; the exceptional-prime rank-three reduction (Theorem 4.3); and the binary higher-valuation metabolicity theorem (Theorem 4.5). We next show what fixed parameters do and do not determine—a valuation-two formula (Theorem 5.1) and a same-parameter separation of two STS(15) (Theorem 5.2)—prove a local Pasch transport law (Theorem 6.1), and record geometric applications. Three supporting developments are placed in appendices: the complete STS(15) defect spectrum (Theorem A.1), the elementary realization theorem (Theorem B.1), and a worked SQS(8) example (Appendix C) with the reproducible finite controls.

Relation to prior work; no firstness. The higher-star diagonal form of t-vs-k subset incidence is due to Wilson [1]; the universal additive/incidence group of a design and its $t = 2$ specialization are due to Caggegi–Falcone–Pavone [2]; the complete invariant factors of point/subspace incidence in $\mathrm{PG}(n, q)$ and $\mathrm{AG}(n, q)$ are due to Chandler–Sin–Xiang [6] (for the general Smith-form landscape of incidence matrices see Sin's survey [5]); the Miquelian and even-order Möbius incidence Smith forms are due to Ringe [23]; the theory of finite symmetric bilinear forms is due to Wall [7] and the discriminant-form/overlattice framework to Nikulin [8]; exact existence of $S(2, 5, v)$ carriers is Hanani's [11]; the Hermitian unital incidence divisors are Hiss's [22]; the STS(15) code/lattice classification is Solé's [14]; and the order-16 Hadamard/rank data are Assmus–Key's [20] (with the five STS(15) binary ranks also in Tonchev–Weishaar [15]). We claim no priority for any of these. The additive/zero-sum embedding theory initiated by [2] has since been extended substantially (e.g. the additivity of symmetric and subspace 2-designs, Buratti–Nakić [3], and the general theory of additive combinatorial designs, Buratti–Merola–Nakić [4]); we use only its basic augmentation setup and assert nothing new on the additive side. The nearest classical framework is the integral/null-design module theory of the *complete* subset design, developed by Graver–Jurkat [18] and Graham–Li–Li [19] and used in the recent integral-design literature (e.g. Keevash [17]); the object here is instead the point-column lattice of a *selected* design—a different lattice—and Theorem 2.1 concerns it, not the complete carrier. The contribution is the selected-design Δ package built on the imports

above; a dated search located no exact prior equivalent of the combined selected-design statement, but the original Wilson and Graver–Jurkat texts were not inspected in full, so this is a negative search with an access caveat, not a firstness certificate.

2 The higher-star defect

For $1 \le r < t$ let λ_r be the number of blocks on a fixed r-subset, and put $d = \gcd\{\lambda_r - \lambda_{r+1} : 1 \le r < t\}$.

Theorem 2.1. *With the notation above,*

$$d\,H_k \;\le\; L \;\le\; H_k.$$

Consequently $\Delta(D) = H_k/L$ is finite, is killed by d, is generated by at most $v-1$ elements, and has order dividing $d^{\,v-1}$. There is a canonical exact sequence

$$0 \to \Delta(D) \to \operatorname{coker}(A) \to \mathbb{Z}/k \to 0,$$

and, for the retained-apex matrix $A^+ = \left[\begin{smallmatrix} \mathbf{1}_P^T \\ A \end{smallmatrix}\right]$, a canonical equality $\Delta(D) = \operatorname{Tor}\operatorname{coker}(A^+)$; the splitting $\operatorname{coker}(A^+) \cong \mathbb{Z} \oplus \Delta(D)$ holds noncanonically.

Proof. Column weight k gives $L \le H_k$. For the lower bound, let u_R be the sum of all block columns containing a fixed r-set R; coordinate counting gives $u_R = \lambda_{r+1}\mathbf{1}_P + (\lambda_r - \lambda_{r+1})\mathbf{1}_R$. Hence, for distinct points x, y and an $(r-1)$-set S disjoint from them,

$$u_{S\cup\{x\}} - u_{S\cup\{y\}} = (\lambda_r - \lambda_{r+1})(e_x - e_y) \in L,$$

and Bézout over $1 \le r < t$ yields $d(e_x - e_y) \in L$ for all x, y. Fix a block B and a point $o \in B$; since $d\,\mathbf{1}_B \in L$,

$$d\,\mathbf{1}_B - \sum_{x \in B\setminus\{o\}} d(e_x - e_o) = dk\,e_o \in L.$$

The vectors $\{e_x - e_o : x \ne o\}$ together with $k\,e_o$ generate H_k, and both $d(e_x - e_o)$ and $dk\,e_o$ lie in L, so $dH_k \le L$. Thus $\Delta(D)$ is finite and killed by d. In H_k/L the block relation $\mathbf{1}_B = k\,e_o + \sum_{x\in B\setminus\{o\}}(e_x - e_o)$ vanishes, expressing the class of $k\,e_o$ through the $v-1$ point-difference classes; hence $\Delta(D)$ is a quotient of $(\mathbb{Z}/d)^{\,v-1}$, so it has at most $v-1$ generators and $|\Delta(D)|$ divides $d^{\,v-1}$. The coordinate-sum map $\operatorname{coker}(A) \to \mathbb{Z}/k$, $[z] \mapsto \sum z \bmod k$, has kernel $H_k/L = \Delta(D)$ and is onto because $[e_x] \mapsto 1$ generates $\mathbb{Z}/k$; and the augmentation identity $\mathbf{1}_P^T A = k\,\mathbf{1}_B^T$, via the one-row retained-apex lemma, gives $\operatorname{coker}(A^+) \cong \mathbb{Z} \oplus \Delta(D)$ noncanonically, with $\Delta(D)$ its torsion. □

The star mechanism (the identity producing the coefficients $\lambda_r - \lambda_{r+1}$) is Wilson's [1]; the universal group into which $\operatorname{coker}(A)$ maps, and the $t = 2$ case, are [2]. The retained-row passage is elementary.

Corollary 2.2. *If $d = 1$ then $L = H_k$ and $\Delta(D) = 0$, i.e. $\operatorname{coker}(A) \cong \mathbb{Z}/k$ and the full incidence Smith form is $1^{\,v-1}k$.*

3 Pointed presentation and the parameter envelope

Fix $x_0 \in P$. Deleting the row x_0 from A and appending the all-ones block row gives a full-row-rank $v \times b$ matrix C_{x_0}; the basis $\{e_x - e_{x_0}\} \cup \{k e_{x_0}\}$ of H_k turns each incidence column into its coordinates away from x_0 followed by a final 1.

Proposition 3.1. *$\Delta(D) \cong \operatorname{coker}(C_{x_0})$, and different choices of x_0 differ by unimodular row operations. The Gram matrix $G = C_{x_0} C_{x_0}^T$ depends only on the ordinary 2-design parameters,*

$$G = \begin{pmatrix} \alpha I_n + \lambda J_n & r\,\mathbf{1}_n \\ r\,\mathbf{1}_n^T & b \end{pmatrix}, \qquad \det G = \alpha^{\,n}\, r/k,$$

so $|\Delta(D)|^2$ divides $\alpha^n r/k$, and for every prime p, $2\,\ell_p(\Delta) \le n\,v_p(\alpha) + v_p(r) - v_p(k)$. Writing $R = \operatorname{im}(C_{x_0}^T)$ and $\bar{R} = (R \otimes \mathbb{Q}) \cap \mathbb{Z}^b$, the group $\bar{R}/R$ is canonically Pontryagin dual to $\Delta(D)$ and embeds in $\operatorname{coker}(G) \cong R^{\#}/R$ as a totally isotropic subgroup for the bilinear discriminant form $\beta_G(z,w) = z^T G^{-1} w \bmod \mathbb{Z}$. (No quadratic refinement is asserted without parity hypotheses.)

Proof. The pointed basis identifies H_k/L with $\operatorname{coker}(C_{x_0})$; the Gram computation is the standard 2-design identity, and $\det G = \alpha^n r/k$ follows from $(rk-\lambda)b - nr^2 = \alpha r/k$. Since C_{x_0} has full row rank v, its finite cokernel has order $|\Delta|$ equal to the greatest common divisor of the $v \times v$ minors of C_{x_0} (the top determinantal divisor d_v). By the Cauchy–Binet formula, $\det G = \det(C_{x_0} C_{x_0}^T)$ equals the sum of the squares of those maximal minors; each minor is divisible by $|\Delta|$, so every summand is divisible by $|\Delta|^2$, whence $|\Delta|^2 \mid \det G$. Taking p-adic valuations gives the primewise bound $2\ell_p(\Delta) \le v_p(\det G) = n\,v_p(\alpha) + v_p(r) - v_p(k)$. The perfect pairing on $\bar{R}/R$ and its identification with $\Delta^\vee$ are overlattice duality [8]; isotropy is $\beta_G|_{\bar{R}/R} = 0$. □

Thus the abstract annihilation of Theorem 2.1 becomes a computable structural invariant: a finite bilinear form determined by the parameters, with the dual defect sitting inside it isotropically.

4 The discriminant form prime by prime

Throughout this section $E_p = \operatorname{coker}(G)_p$ with the pairing β_G, and ℓ_p denotes length. The maximal totally isotropic subgroup of E_p bounds Δ_p from above; the square-order bound of Proposition 3.1 is the statement that this maximum is at most $|E_p|^{1/2}$.

4.1 Good primes

Proposition 4.1 (good-prime decomposition)**.** *For every prime $p \nmid n$, the pointed lattice splits β_G-orthogonally as*

$$E_p \;\cong\; \big(W_p/\alpha W_p,\ (x,y) \mapsto x\cdot y/\alpha\big) \;\perp\; \big(\operatorname{coker} K_p,\ K_p^{-1}\big), \quad K_p = \begin{pmatrix} n(rk-\lambda) & nr \\ nr & b \end{pmatrix},$$

where $W_p = \{x \in \mathbb{Z}_p^n : \mathbf{1}^T x = 0\}$; here $\det K_p = n\alpha r/k$.

Proof. Since $p \nmid n$, the vector $\mathbf{1}_n$ has unit norm n, so $\mathbb{Z}_p^n = W_p \perp \mathbb{Z}_p \mathbf{1}_n$; adjoining the apex coordinate $\mathbb{Z}_p e_*$ gives the orthogonal splitting $\mathbb{Z}_p^v = W_p \perp T_p$ with $T_p = \mathbb{Z}_p \mathbf{1}_n \oplus \mathbb{Z}_p e_*$. Restricting G from Proposition 3.1: on W_p (coordinate sum zero) it is α times the dot product, which is unimodular there; on T_p it is the block K_p, whose determinant $n\alpha r/k$ follows from $(rk-\lambda)b - nr^2 = \alpha r/k$. Passing to cokernels and inverse-Gram pairings in this orthogonal basis gives the stated sum; ℓ_p of each summand is read from the entry valuations of K_p. □

Theorem 4.2 (odd valuation-one Witt obstruction)**.** *Let p be odd with $p \nmid k\lambda$ and $v_p(\alpha) = 1$, and suppose $n = v - 1$ is even, $n = 2m$. Then $E_p \cong \mathbb{F}_p^n$ carries a nondegenerate symmetric form, its dual selected defect is totally isotropic, and its minimal number of generators $\mu(\Delta_p)$ satisfies*

$$\mu(\Delta_p) \le m \quad \text{if } (-1)^m k \text{ is a square mod } p, \qquad \mu(\Delta_p) \le m-1 \quad \text{otherwise.}$$

The second line is one generator stronger than $|\Delta_p|^2 \mid \det G$. For $v = 5 + 60t$ with $t \ge 1$ and $3 \nmid t$, Hanani's theorem [11] supplies an $S(2, 5, v)$; then $n = 4+60t$ is even, at $p = 3$ one has $m = 2+30t$, $(-1)^m k = 5$ a nonsquare, so $\Delta_3 \cong (\mathbb{Z}/3)^\mu$ with $\mu \le 1 + 30t$, whereas the coarse envelope permits $2 + 30t$. This is an infinite strict obstruction.

Proof. Under the hypotheses $G \equiv \lambda J_v \pmod p$ has rank one, and $v_p(\det G) = n$ forces every nonunit invariant factor to have valuation one, so $E_p \cong \mathbb{F}_p^n$; the scaled form $p\,\beta_G$ has determinant square class k/u^n ($u = \alpha/p$), equal to k when n is even. By the classification of nondegenerate symmetric bilinear forms over a finite field [10], a $2m$-dimensional such space has Witt index m iff $(-1)^m$ times its determinant is a square, else $m-1$; the dual defect is a totally isotropic subspace, so its dimension is at most the Witt index. The $S(2, 5, v)$ evaluation is direct. □

4.2 Exceptional primes and the binary boundary

For $p \mid n$ the root/line split fails; a different primitive orthogonal splitting survives.

Theorem 4.3 (exceptional-prime reduction)**.** *For every prime $p \mid n$ the pointed lattice is $\mathbb{Z}_p$-congruent to $\alpha D_{n-2} \perp C_p$, where $D_{n-2} = I + J$ has rank $n-2$ and determinant $n-1$ (a p-adic unit, so D_{n-2} is unimodular over $\mathbb{Z}_p$), and*

$$C_p = \begin{pmatrix} \alpha n/(n-1) & -\alpha/(n-1) & 0 \\ -\alpha/(n-1) & \lambda + \alpha/(n-1) & r \\ 0 & r & b \end{pmatrix}, \qquad \det C_p = \frac{\alpha^2 r}{k(n-1)}.$$

Hence all but at most three local invariant factors have valuation $v_p(\alpha)$, and the remaining three are read from the determinantal ideals of C_p. At $p = 2$ the root block αD_{n-2} is type II; if $v_2(\alpha) = 1$ and $k\lambda$ is odd, the whole discriminant space is symplectic of dimension n, so bilinear isotropy meets—but does not beat—the square-order bound.

Proof. Put $t = n - 1$, a p-adic unit. The sublattice $U = \{x \in \mathbb{Z}_p^n : x_n = 0,\ \sum_i x_i = 0\}$ has dot-product Gram $D_{n-2} = I + J$ ($\det = n-1$, unimodular over $\mathbb{Z}_p$) and is G-orthogonal to $s = (\mathbf{1}_n - e_n)/(n-1)$, $a = e_n - s$, and the apex e_*. A direct computation gives $\langle a, a\rangle_G = \alpha n/(n-1)$, $\langle a, s\rangle_G = -\alpha/(n-1)$, $\langle s, s\rangle_G = \lambda + \alpha/(n-1)$, $\langle a, e_*\rangle_G = 0$, and $\langle s, e_*\rangle_G = r$, so in the local basis (a, s, e_*) the residual Gram is exactly C_p; the determinant identity $\alpha^{n-2} \det D_{n-2} \det C_p = \alpha^n r/k = \det G$ together with adjugation gives $\det C_p$ and C_p^{-1}, and the three residual exponents are the successive determinantal-ideal valuations. This is a unimodular local change of basis, not the invalid zero-sum-root/constant-line split (whose index is divisible by p). At $p = 2$, $D_{n-2} = I + J$ has even diagonal and odd determinant, hence gives a type-II $2^{v_2(\alpha)}$-modular constituent (standard $\mathbb{Z}_2$ Jordan splitting [9]); in the valuation-one unit case $2G^{-1} \bmod 2$ is alternating, so the discriminant space is symplectic and its maximal isotropic dimension equals the square-order bound. □

The remaining binary regime is higher valuation, $A := v_2(\alpha) \ge 2$. Here the growing constituent is the type-II 2^A-modular group $(\mathbb{Z}/2^A)^{n-2}$.

Lemma 4.4 (plateau metabolicity)**.** *Every homogeneous type-II 2^A-modular symmetric bilinear plane $U(2^A)$ or $V(2^A)$ is metabolic for all $A \geq 1$; consequently a type-II 2^A-modular form of even rank is metabolic and carries a half-order isotropic subgroup.*

Proof. For $U(2^A) = 2^{-A}\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ the cyclic group $\langle e_1 \rangle$ is a Lagrangian. For $V(2^A) = 2^{-A}\left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, take $S = \langle (2^c, 0), (0, 2^c) \rangle$ when $A = 2c$ and $S = \langle (2^c, 0), (0, 2^{c+1}) \rangle$ when $A = 2c + 1$; a direct check gives $\beta|_S \equiv 0$ and $|S| = 2^A$. A sum of metabolic forms is metabolic (sum of Lagrangians). This is a specialization of Wall's classification [7]. □

Theorem 4.5 (binary higher-valuation confinement)**.** *Let $A = v_2(\alpha) \geq 2$ and $2 \mid n$. In the decomposition $E_2 \cong (\mathbb{Z}/2^A)^{n-2} \perp \operatorname{coker}(C_2)$ of Theorem 4.3, the root plateau is metabolic (Lemma 4.4), so*

$$\operatorname{maxiso}(E_2) \;\geq\; 2^{A(n-2)/2} \cdot \operatorname{maxiso}(\operatorname{coker} C_2),$$

and the isotropy defect of E_2 is confined to the explicit rank-≤ 3 residual block C_2, bounded by $(2A + v_2(r) - v_2(k))/2$, independently of $n = v - 1$. In particular there is no binary analogue of the n-growing strict family of Theorem 4.2: at fixed parameter valuations the bilinear form loses at most a bounded amount, determined by a finite Jordan case split of C_2.

Proof. Orthogonality is Theorem 4.3. If M is metabolic with Lagrangian L_M then $\operatorname{maxiso}(M \perp N) \geq |L_M| \operatorname{maxiso}(N)$ (take $L_M \perp T$ for $T \leq N$ isotropic), giving the displayed bound with M the plateau. Hence $|E_2|^{1/2}/\operatorname{maxiso}(E_2) \leq |\operatorname{coker} C_2|^{1/2}$, and $\ell_2(\operatorname{coker} C_2) = v_2(\det C_2) = 2A + v_2(r) - v_2(k)$ by Theorem 4.3, all independent of n. □

Theorem 4.5 extends the valuation-one symplectic no-loss statement of Theorem 4.3 to every higher valuation and *reduces* the remaining binary obstruction to the explicit rank-≤ 3 residual block C_2—a finite, n-independent case split, not a full classification of that block; the equality $\operatorname{maxiso}(E_2) = 2^{A(n-2)/2}\operatorname{maxiso}(\operatorname{coker} C_2)$ can fail, so only the displayed one-sided bound is used.

5 Fixed parameters: determinacy and its failure

The pointed invariant refines the ordinary parameters. Two fixed-parameter facts frame how much: at valuation two the local defect is pinned by a single modular rank, yet at valuation one the ordinary parameters—even the whole Gram envelope—do not determine Δ at all.

Theorem 5.1 (symmetric valuation two)**.** *Let D be symmetric of order $w = k - \lambda$ (distinct from the global $n = v - 1$) and $\rho_p = \operatorname{rank}_{\mathbb{F}_p}(C_{x_0})$. If $v_p(w) = 2$ then*

$$\Delta(D)_p \;\cong\; (\mathbb{Z}/p)^{\,v+1-2\rho_p} \;\oplus\; (\mathbb{Z}/p^2)^{\,\rho_p - 1},$$

so the intrinsic local exponent is p^2 exactly when $\rho_p \geq 2$. For the calibration $\mathrm{PG}(2, p^2)$ this recovers the Chandler–Sin–Xiang value [6]; the exact $\mathrm{PG}(2, 4)$ pointed form is $1^{10}2^2 4^9$.

Proof. For a symmetric design $|\Delta|^2 = (k - \lambda)^{v-1} = w^{v-1}$, so $v_p(|\Delta_p|) = (v - 1)v_p(w)/2 = v - 1$; as $v_p(w) = 2$, Δ_p has exponent dividing p^2, say $\Delta_p \cong (\mathbb{Z}/p)^a \oplus (\mathbb{Z}/p^2)^b$. The mod-$p$ reduction of $\operatorname{coker}(C_{x_0})$ has dimension $v - \rho_p$, so $a + b = v - \rho_p$, while $v_p(|\Delta_p|) = a + 2b = v - 1$. Solving gives $b = \rho_p - 1$ and $a = v + 1 - 2\rho_p$. The $\mathrm{PG}(2, p^2)$ multiplicities are the Chandler–Sin–Xiang invariant factors [6]. □

Theorem 5.2 (same-parameter separation)**.** *The point/line design of* $\mathrm{PG}(3,2)$ *and the Bose averaging system on* $\mathbb{Z}_5 \times \mathbb{Z}_3$ *are both simple* 2-$(15,3,1)$ *designs with the same parameter Gram matrix, yet*

$$\Delta(\mathrm{PG}(3,2)) \cong (\mathbb{Z}/2)^4, \qquad \Delta(\mathit{Bose}) = 0,$$

with full incidence Smith forms $1^{11}2^3 6$ *and* $1^{14}3$*. Hence ordinary parameters, and even the parameter Gram envelope, do not determine* Δ*.*

Proof. Both constructions produce 35 triples covering the 105 pairs once. As $6\Delta = 0$ with 6 squarefree, the modular ranks $\mathrm{rank}_{\mathbb{F}_2}(C_{x_0})$ and $\mathrm{rank}_{\mathbb{F}_3}(C_{x_0})$ determine Δ; they are $11, 15$ for $\mathrm{PG}(3,2)$ and $15, 15$ for Bose. The finite verification is reproduced by the included script (Appendix D). □

6 Local transport

Theorem 6.1 (Pasch transport)**.** *Let* D, D' *differ by a single Pasch (quadrilateral) switch. The two* local four-block lattices*—spanned by the four old, resp. four new, blocks that change—agree at every odd prime, and localized at* 2 *each has index two in their common saturation. Consequently every odd-primary part of* Δ *is preserved, the defect-order ratio* $|\Delta(D')|/|\Delta(D)|$ *lies in* $\{1/2, 1, 2\}$ *(so one of* $\Delta(D), \Delta(D')$ *may already equal the other), and the minimum numbers of binary generators of* $\Delta(D)$ *and* $\Delta(D')$ *differ by at most one.*

Proof. Only the eight blocks meeting the switched quadrilateral change. Let L_0 be the sublattice spanned by the unaffected blocks and M_O, M_N the *local* lattices spanned by the four old, resp. new, blocks. At every odd prime $M_O = M_N$; localized at $p = 2$ each of M_O, M_N has index two in their common saturation $\widehat{M}$, so the local exchange groups $E_O = \widehat{M}/M_O$ and $E_N = \widehat{M}/M_N$ are both $\mathbb{Z}/2$, read off from the determinantal divisors of the four-block minors. The *global* defects are $\Delta(D) = H_k/(L_0 + M_O)$ and $\Delta(D') = H_k/(L_0 + M_N)$, formed only after adjoining the unchanged L_0; both surject onto the common quotient $H_k/(L_0 + \widehat{M})$ with kernel a quotient of E_O, resp. E_N. Hence every odd-primary part is preserved, the defect-order ratio lies in $\{1/2, 1, 2\}$, and the minimum binary generator numbers differ by at most one. □

7 Geometric applications

Finite Möbius planes. A finite Möbius plane is a 3-$(q^2+1, q+1, 1)$ design; its star differences are $q^2 - 1$ and q, with gcd = 1, so $d = 1$ and Corollary 2.2 gives the full incidence Smith form $1^{q^2}(q+1)$ and $\Delta = 0$. The Miquelian instance and the arbitrary even-order case are Ringe's [23]; this is the vanishing corollary of Theorem 2.1, not a new Möbius theorem.

Hermitian unitals. The classical Hermitian unital $H(q)$—the 2-$(q^3+1, q+1, 1)$ design of Key's explicit finite-field/secant-line construction [21]—has, by Hiss's incidence divisors [22], the incidence cokernel

$$\mathrm{coker}(A(H(q))) \cong (\mathbb{Z}/(q+1))^{q^2-q+1}.$$

Removing the augmentation factor via Theorem 2.1 yields the homocyclic family

$$\Delta(H(q)) \cong (\mathbb{Z}/(q+1))^{q^2-q},$$

so $\Delta(H(4)) = (\mathbb{Z}/5)^{12}$. Hiss owns the Smith row and Key the construction; this is a supporting application.

A nonzero $t = 3$ instance—the design $\mathrm{SQS}(8) = \mathrm{AG}(3,2)$, for which $d = 2$ and $\Delta \cong (\mathbb{Z}/2)^3$—is worked out in Appendix C, where it shows that the hypothesis $d = 1$ of Corollary 2.2 has content.

A The STS(15) defect spectrum

This appendix records the complete realized defect spectrum on 15 points, the general squarefree trichotomy, and the ternary augmentation splitting for triple designs. The first two are catalog-level consequences of the main results together with imported p-ranks; the third is proved here and is used by the ternary case of Appendix B.

Theorem A.1 (the STS(15) spectrum)**.** *For a Steiner triple system on* 15 *points,* $\operatorname{rank}_{\mathbb{F}_3}(A) = 14$ *and* $\operatorname{rank}_{\mathbb{F}_3}(C_{x_0}) = 15$*; hence* $\Delta = (\mathbb{Z}/2)^a$ *with* $a = 15 - \operatorname{rank}_{\mathbb{F}_2}(A)$*. The realized binary ranks are* $11, 12, 13, 14, 15$ *[20], so the complete realized spectrum is*

$$\Delta \in \{0,\ \mathbb{Z}/2,\ (\mathbb{Z}/2)^2,\ (\mathbb{Z}/2)^3,\ (\mathbb{Z}/2)^4\},$$

with full incidence forms $1^{14}3$, $1^{14}6$, $1^{13}2{\cdot}6$, $1^{12}2^26$, $1^{11}2^36$ *respectively. More generally, using that a nontrivial* STS(v) *has full p-rank for every prime* $p \geq 5$ *[12] together with the no-simultaneous-deficiency theorem of Shi–Xu–Krotov [13], every nontrivial* STS *defect is a pure* 2*-group, a pure* 3*-group, or zero.*

The STS(15) ternary statement is structural (a nonconstant ternary dual vector would partition the points into three STS(5)'s, impossible); the five binary ranks are imported from Assmus–Key [20] and Tonchev–Weishaar [15], and the code/affine-lattice classification of the 80 systems is Solé's [14]. The general trichotomy rests on the cited p-rank inputs; no firstness attaches to the squarefree translation.

Proposition A.2 (ternary augmentation splitting)**.** *For every nontrivial* 2-$(v, 3, \lambda)$ *design with* $v > 3$*, the all-ones block row is not in the* $\mathbb{F}_3$*-row span of the incidence matrix* $A\colon \mathbb{Z}^{\mathcal{B}} \to \mathbb{Z}^P$*; consequently the augmentation* $\varepsilon\colon \operatorname{coker}(A) \to \mathbb{Z}/3$ *splits, so* $\operatorname{coker}(A) \cong \Delta(D) \oplus \mathbb{Z}/3$.

Proof. Suppose $a \in \mathbb{F}_3^P$ satisfies $\sum_{x \in B} a_x = 1$ for every block B, and partition the points by fibre $P_i = \{x : a_x = i\}$, $n_i = |P_i|$ $(i \in \mathbb{Z}/3)$. A triple sums to 1 in $\mathbb{F}_3$ only if its values form the multiset $(i, i, i+1)$; call such a block *type* i. Each pair inside P_i lies in λ blocks and forces the third point into P_{i+1}, giving $\lambda\binom{n_i}{2}$ blocks of type i; counting the same blocks by their two P_i–P_{i+1} cross-pairs gives $2 \cdot (\#\,\text{type } i) = \lambda\, n_i n_{i+1}$. Hence $\lambda\, n_i(n_i - 1) = \lambda\, n_i n_{i+1}$, so $n_{i+1} = n_i - 1$ whenever $n_i > 0$. Applied cyclically over $i \in \mathbb{Z}/3$ this forces three strict decreases $n_0 > n_1 > n_2 > n_0$, impossible unless some fibre is empty; and with an empty fibre the surviving instances of $n_{i+1} = n_i - 1$ bound $n_0 + n_1 + n_2 \leq 3$ (the maximum is the rotation $(0, 2, 1)$), contradicting $v > 3$. Thus no such a exists, so the all-ones row lies outside the $\mathbb{F}_3$-row span of A. Row-space duality then supplies $y \in \mathbb{F}_3^{\mathcal{B}}$ with $Ay = 0$ in $\mathbb{F}_3^P$ and $\mathbf{1}_{\mathcal{B}}^T y = 1$. Lift y to $\tilde{y} \in \mathbb{Z}^{\mathcal{B}}$; then $A\tilde{y} \equiv 0 \pmod 3$, so $z := A\tilde{y}/3 \in \mathbb{Z}^P$ is integral, its class $[z] \in \operatorname{coker}(A)$ satisfies $3[z] = [A\tilde{y}] = 0$, and, using $\mathbf{1}_P^T A = 3\,\mathbf{1}_{\mathcal{B}}^T$ (each block has three points),

$$\varepsilon([z]) = \mathbf{1}_P^T z \bmod 3 = \tfrac{1}{3}\mathbf{1}_P^T A\tilde{y} \bmod 3 = \mathbf{1}_{\mathcal{B}}^T \tilde{y} \bmod 3 = 1.$$

Hence $\mathbb{Z}/3 \to \operatorname{coker}(A)$, $1 \mapsto [z]$, is a section of ε, and $\operatorname{coker}(A) \cong \Delta(D) \oplus \mathbb{Z}/3$. □

B Realization of elementary defects

Theorem B.1 (elementary realization)**.** *For every fixed* $a \geq 1$*, all sufficiently large* n *admit an* STS$(2^n - 1)$ *with* $\Delta \cong (\mathbb{Z}/2)^a$*; for every fixed* $b \geq 1$*, all sufficiently large* n *admit an* STS(3^n) *with* $\Delta \cong (\mathbb{Z}/3)^b$.

Proof. By Theorem A.1, $\Delta = (\mathbb{Z}/2)^a$ is exactly a binary rank deficiency a of the incidence matrix, and $(\mathbb{Z}/3)^b$ a ternary deficiency b; realizing the group is realizing the rank. Jungnickel and Tonchev [16] enumerate the Steiner triple systems with classical parameters and prescribed 2-rank (resp. 3-rank), and show that for each fixed deficiency the number of such systems grows without bound as the order increases. Hence, for every fixed a (resp. b), systems of exactly that deficiency exist for all sufficiently large n; valuation one, the STS primary purity of Theorem A.1, and the ternary augmentation splitting of Proposition A.2 then convert the deficiency into the stated homocyclic group. The prescribed-rank carriers and their counts are Jungnickel–Tonchev's; only the translation into Δ is added here. □

C A nonzero example: SQS(8)

The Steiner quadruple system $\mathrm{SQS}(8) = \mathrm{AG}(3,2)$ is a 3-$(8,4,1)$ design: its 14 blocks are the affine planes (cosets of the seven 2-dimensional subspaces) of $\mathbb{F}_2^3$. Its star sequence has $\lambda_1 = 7$, $\lambda_2 = 3$, $\lambda_3 = 1$, so the star differences are 4 and 2 and $d = \gcd\{4,2\} = 2 \neq 1$; Corollary 2.2 therefore does *not* force $\Delta = 0$. The pointed presentation is 8×14, and enumerating its $\binom{14}{8} = 3003$ maximal 8×8 minors gives

$$\Delta(\mathrm{SQS}(8)) \cong (\mathbb{Z}/2)^3, \qquad \text{full incidence Smith form } 1^4 2^3 4.$$

Here SQS(8) is the smallest nontrivial Steiner quadruple system, and this nonzero defect shows that the hypothesis $d = 1$ of Corollary 2.2 has genuine content: dropping it admits a nontrivial 2-group. The computation is reproduced from scratch by the verifier (Appendix D).

D Reproducibility of the finite controls

The self-contained finite controls—the SQS(8) defect $(\mathbb{Z}/2)^3$ with form $1^4 2^3 4$; the same-parameter PG$(3,2)$/Bose separation, forms $1^{11}2^3 6$ and $1^{14}3$ with defects $(\mathbb{Z}/2)^4$ and 0; and the smallest Möbius plane $M(2) =$ 3-$(5,3,1)$ with form $1^4 3$—are reproduced from scratch by `scripts/verify.py`, which computes each Smith form with `python-flint` and cross-checks the binary and ternary ranks by an independent modular route; its certificate is byte-identical under `python` and `python -O`. The other finite forms cited in the text (the Möbius ladder for $q \geq 3$ [23], the PG$(2,4)$ valuation-two form and the general projective calibration [6], the Hermitian divisors [22], and the intermediate STS(15) ranks obtained by Pasch transport of Theorem 6.1) are imported or follow from the stated results, not recomputed here. The general theorems of the main text—the sandwich, the pointed presentation, the prime-by-prime analysis, the transport law, and the Möbius/Hermitian applications—use no finite census; the finite inputs are confined to the same-parameter separation (Theorem 5.2) and the worked SQS(8) example (Appendix C).

Data and Code Availability

The companion repository and its archived v1.0.0 release contain this LaTeX source (main text and appendices), the compiled title-named PDF, and a single self-contained verifier `scripts/verify.py` that constructs the finite controls from scratch with no external data—the SQS(8) defect $(\mathbb{Z}/2)^3$, the same-parameter PG$(3,2)$/Bose separation, and the Möbius plane $M(2)$—computing each Smith form with `python-flint`, enforcing an independent modular cross-check, and asserting byte-identity against the frozen certificate `certificates/expected_verification.json`. The

output is byte-identical under `python` and `python -O`. Classical inputs are attributed in `docs/SOURCE_LOCK.md` and the bibliography; the claim ledger and scope boundary live in `docs/`. See `REPRODUCE.md`.

References

[1] R. M. Wilson, *A diagonal form for the incidence matrices of t-subsets vs. k-subsets*, European J. Combin. **11** (1990), 609–615, DOI 10.1016/S0195-6698(13)80046-7.

[2] A. Caggegi, G. Falcone, M. Pavone, *On the additivity of block designs*, J. Algebraic Combin. **45** (2017), 271–294. DOI 10.1007/s10801-016-0707-5.

[3] M. Buratti, A. Nakić, *Additivity of symmetric and subspace* 2*-designs*, Des. Codes Cryptogr. (2024), 3561–3572, DOI 10.1007/s10623-024-01452-4.

[4] M. Buratti, F. Merola, A. Nakić, *Additive combinatorial designs*, Des. Codes Cryptogr. **93** (2025), no. 7, 2717–2740, DOI 10.1007/s10623-025-01594-z.

[5] P. Sin, *Smith normal forms of incidence matrices*, arXiv:1401.8210.

[6] D. B. Chandler, P. Sin, Q. Xiang, *The invariant factors of the incidence matrices of points and subspaces in $PG(n, q)$ and $AG(n, q)$*, Trans. Amer. Math. Soc. **358** (2006), 4935–4957.

[7] C. T. C. Wall, *Quadratic forms on finite groups, and related topics*, Topology **2** (1963), 281–298.

[8] V. V. Nikulin, *Integral symmetric bilinear forms and some of their applications*, Math. USSR-Izv. **14** (1980), 103–167.

[9] J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren **290**, Springer, 1999.

[10] K.-U. Schmidt, *Symmetric bilinear forms over finite fields with applications to coding theory*, arXiv:1410.7184.

[11] H. Hanani, *On balanced incomplete block designs with blocks having five elements*, J. Combin. Theory Ser. A **12** (1972), 184–201.

[12] J. Doyen, X. Hubaut, M. Vandensavel, *Ranks of incidence matrices of Steiner triple systems*, Math. Z. **163** (1978), 251–259.

[13] M. Shi, Y. Xu, D. Krotov, *The number of the non-full-rank Steiner triple systems*, J. Combin. Des. **27** (2019), DOI 10.1002/jcd.21663.

[14] P. Solé, *Codes, lattices and Steiner systems*, Electron. J. Combin. **4** (1997), no. 1, #R6.

[15] V. D. Tonchev, R. S. Weishaar, *Steiner triple systems of order* 15 *and their codes*, J. Statist. Plann. Inference **58** (1997), 207–216.

[16] D. Jungnickel, V. D. Tonchev, *Counting Steiner triple systems with classical parameters and prescribed rank*, J. Combin. Theory Ser. A **162** (2019), 10–33. arXiv:1709.06044.

[17] P. Keevash, *The existence of designs*, arXiv:1401.3665.

[18] J. E. Graver, W. B. Jurkat, *The module structure of integral designs*, J. Combin. Theory Ser. A **15** (1973), 75–90.

[19] R. L. Graham, S.-Y. R. Li, W.-C. W. Li, *On the structure of t-designs*, SIAM J. Algebraic Discrete Methods **1** (1980), 8–14.

[20] E. F. Assmus, Jr., J. D. Key, *Hadamard matrices and their designs: a coding-theoretic approach*, Trans. Amer. Math. Soc. **330** (1992), 269–293.

[21] J. D. Key, *Some applications of Magma in designs and codes: oval designs, Hermitian unitals and generalized Reed–Muller codes*, J. Symbolic Comput. **31** (2001), 37–53, DOI 10.1006/jsco.1999.1007.

[22] G. Hiss, *Hermitian function fields, classical unitals, and representations of 3-dimensional unitary groups*, Indag. Math. (N.S.) **15** (2004), 223–243, DOI 10.1016/S0019-3577(04)90016-X.

[23] C. Ringe, *Über die Elementarteiler der Inzidenzmatrix des Ree-Unitals*, Dissertation, RWTH Aachen, 2008; the Möbius/inversive-plane incidence Smith forms are in Ch. 8.